

Pediatric Pharmacotherapy

Introduction

- Pediatric clinical practice focuses on care for infants, children, and adolescents.
- Goal: Optimize health, growth, and development toward adulthood.
- Clinicians advocate for this vulnerable population to ensure their well-being.
- Care occurs in inpatient and outpatient settings, requiring additional considerations for drug selection and monitoring.

Fundamentals of Pediatric Patients

- **Age Classification:**
 - Pediatric patients: Younger than 18 years (may include up to age 21 in some cases).
 - Age expressed in days, weeks, months, and years.
- **Further Classification:**
 - Based on birth weight and prematurity status (refer to Table 2–1 for details).

Table 2–1 Pediatric Age Groups, Age Terminology, and Weight Classification¹⁻³

Age Group	Age
Neonate	≤ 28 days (4 weeks) of life
Infant	29 days to < 12 months
Child	1–12 years
Adolescent	13–17 years (most common definition)
Age Terminology	Definition
GA	Age from date of mother's first day of last menstrual period to date of birth
Full term	Describes infants born at 37-weeks' gestation or greater
Premature	Describes infants born before 37-weeks' gestation
Small for GA	Neonates with birth weight below the 10th percentile among neonates of the same GA
Large for GA	Neonates with birth weight above the 90th percentile among neonates of the same GA
Chronological or postnatal age	Age from birth to present, measured in days, weeks, months, or years
Corrected or adjusted age	May be used to describe the age of a premature child up to 3 years of age: Corrected age = Chronological age in months – [(40 – GA at birth in weeks) × 1 month ÷ 4 weeks]. For example, if a former 29-week GA child is now 10 months old chronologically, his corrected age is approximately 7 months: 10 months – [(40–29 weeks) × 1 month ÷ 4 weeks] = 7.25 months
Weight Classification	Definition
LBW infant	Premature infant with birth weight between 1500 and 2500 g
VLBW infant	Premature infant with birth weight 1000 g to < 1500 g

ELBW infant	Premature infant with birth weight < 1000 g
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ELBW, extremely low birth weight; GA, gestational age; LBW, low birth weight; VLBW, very low birth weight.

Growth and Development

- Children are monitored for:
 - **Physical Development:** Head circumference, weight, length/stature, weight-for-length, BMI.
 - **Motor, Cognitive, and Psychosocial Milestones:** Assessed during routine well-child visits.
- Growth assessment tools:
 - **< 2 years old:** WHO growth standards.
 - **≥ 2 years old:** CDC Growth Charts.
- Growth charts evaluate:
 - Nutritional issues like poor weight/height gain or failure to thrive.
 - Definitions:
 - **Underweight:** BMI < 5th percentile for age.
 - **Overweight:** BMI 85th–94th percentile for age.
 - **Obese:** BMI > 95th percentile for age.
- Special considerations:
 - Charts may not accurately evaluate children with congenital diseases.
 - Disease-specific growth charts (e.g., Turner syndrome, cystic fibrosis) may be used.
- **Physical Development:**
 - Head circumference, weight, length/stature, weight-for-length, BMI.
 - Blood pressure increases with age:
 - Neonates: ~70/50 mmHg.
 - Adolescents: ~110/65 mmHg.
 - Heart rate decreases with age:
 - Neonates/Infants: 90–205 beats/min.
 - By ~10 years: 60–100 beats/min (adult rates).
 - Body temperature:
 - Rectal measurement (most accurate for <3 months).
 - Temporal artery: For ≥3 months.
 - Tympanic: For ≥6 months.
 - Oral: Reliable for ≥4–5 years.
 - Fever definitions:
 - Rectal/Otic/Temporal: ≥100.4°F (38°C).
 - Oral: ≥100°F (37.8°C).
 - Axillary: ≥99°F (37.2°C).
 - Low-grade fever: 100°–102°F (37.8°–38.9°C).
 - Antipyretic treatment: For temperature >38.3°C (101°F) with discomfort.
- **Pain Assessment:**

- Challenges in neonates/infants due to lack of communication.
- Indicators:
 - Physiological: Increased HR, RR, BP, decreased oxygen saturation.
 - Behavioral: High-pitched crying, facial expressions.
- Pain assessment tools:
 - **FACES Scale:** For children ≥ 3 years to self-report pain using facial expressions.
 - **FLACC Scale:** For patients 2 months–7 years or non-communicative; clinician evaluates facial expression, leg movement, activity, crying, consolability.
- **Motor, Cognitive, and Psychosocial Milestones:** Assessed during routine well-child visits.
- **Growth Assessment Tools:**
 - WHO growth standards for <2 years.
 - CDC growth charts for ≥ 2 years.
- **Growth Chart Evaluations:**
 - Identify nutritional issues like poor weight/height gain or failure to thrive.
 - Definitions:
 - Underweight: BMI <5 th percentile for age.
 - Overweight: BMI 85th–94th percentile for age.
 - Obese: BMI >95 th percentile for age.
 - Special Considerations:
 - Charts may not suit children with congenital diseases.
 - Disease-specific growth charts available (e.g., Turner syndrome, cystic fibrosis).

Fluid Requirements

Table 2–3 Maintenance Fluid Calculations by Body Weight¹⁶

Patient Body Weight	Maintenance Fluid Requirement
< 10 kg	100 mL/kg/day
11–20 kg	1000 mL + 50 mL/kg over 10 kg
> 20 kg	1500 mL + 20 mL/kg over 20 kg

Patient Encounter Part 1

HM is a 33-week GA premature newborn boy weighing 1.6 kg (3.5 lb), length 39 cm (15.4 in), born to a 25-year-old woman today. HM is currently admitted to the neonatal intensive care unit.

What is HM's weight classification as a neonate?

Calculate HM's corrected age for 6 months from today.

How much maintenance fluid per day (mL) and overall rate (mL/hour), is appropriate at this time for HM?

Effects of Pharmacokinetic and Pharmacodynamic Differences on Drug Therapy

- **Drug selection strategies:**
 - Differ based on age and disease state due to variations in pathophysiology, pharmacokinetics, and pharmacodynamics between pediatric and adult patients.
 - Pediatric patients often require different medications than adults for the same diseases (e.g., phenobarbital for neonatal seizures vs. newer antiepileptic drugs for adults).
 - Common therapeutic serum drug concentrations may be similar (e.g., vancomycin for children and adults).
- **Factors influencing pediatric drug therapy:**
 - Age, weight, height, disease, comorbidities, developmental pharmacokinetics, and available dosage forms.
 - Pediatric doses are often weight-based (e.g., mg/kg/dose) compared to uniform adult doses (e.g., mg/day).
- **Pediatric dosing considerations:**
 - Accurate weight measurement is crucial for prescribing or dispensing medications.
 - Pediatric doses may exceed adult doses on a per kg basis due to pharmacokinetic and pharmacodynamic differences.
 - Use pediatric drug dosing guides for accurate dosing.
- **Key differences in pediatric and adult pharmacokinetics and pharmacodynamics:**
 - Age-dependent organ development affects drug absorption, distribution, metabolism, and excretion.

Absorption

- **Oral Absorption**
 - Differences due to gastric acid secretion and pancreatic/biliary function.
 - **Gastric pH:** Increased (pH 6–8) due to lower acid output; reaches adult values by ~2 years.
 - Increased serum concentrations of weak bases and acid-labile drugs (e.g., penicillin).
 - Decreased serum concentrations of weak acid drugs (e.g., phenobarbital).
 - **Gastric emptying and intestinal transit time:** Delayed in premature infants, increasing drug contact with GI mucosa.
 - **Impact of diseases:** Conditions like gastroesophageal reflux, respiratory distress syndrome, and congenital heart disease can further delay gastric emptying.
 - **Pancreatic and biliary function:** Reduced in newborns (50% less amylase/lipase secretion than adults); reaches adult levels between 1–5 years.
 - Decreased bioavailability of prodrug esters (e.g., erythromycin).
 - **Drug dosing:** Often extrapolated from adult data due to limited pediatric bioavailability studies.

- **Topical/Percutaneous Absorption**
 - Increased due to thinner stratum corneum, enhanced cutaneous perfusion, and greater body surface-to-weight ratio.
 - Topical medications (e.g., corticosteroids) should be used sparingly to minimize systemic exposure.
- **Intramuscular (IM) Absorption**
 - Erratic in premature/full-term infants due to:
 - Variable perfusion, poor muscle contraction, and reduced muscle mass.
 - Reserved for cases where other routes are inaccessible (e.g., neonatal sepsis with ampicillin and gentamicin).
- **Intrapulmonary Absorption**
 - Influenced by lung anatomy and drug delivery.
 - Smaller airways and lower inspiratory volume concentrate drugs in upper/central airways.
 - Factors affecting absorption: Particle size, breathing pattern, and route (oral vs nasal).
 - Considerations for devices like nebulizers or inhalers.
- **Rectal Absorption**
 - Erratic due to increased peristalsis, leading to early expulsion of dosage forms.
 - Limited use unless other routes are inaccessible (e.g., severe nausea/vomiting, seizures).
 - **Advantages:**
 - Bypasses first-pass metabolism when absorbed through the lower rectum.
 - **Limitations:**
 - Variable availability of rectal dosage forms.
 - High osmolality/large volume may limit use of oral liquids rectally.

Volume of Distribution

- **Apparent Vd in Pediatrics:**
 - Normalized by body weight and expressed as L/kg.
 - Increased extracellular fluid and total body water per kg in neonates/infants lead to:
 - Higher Vd for water-soluble drugs (e.g., aminoglycosides).
 - Decrease with age.
 - Neonates/infants require higher doses by weight (mg/kg) than older children to achieve the same therapeutic serum levels.
- **Factors Affecting Vd:**
 - **Fluid Overload and Diuresis:** Can alter Vd; must be evaluated during dosing.
 - **Extracorporeal Membrane Oxygenation (ECMO):**
 - Adds volume to the circuit and causes potential fluid changes (e.g., edema).
 - Therapeutic drug monitoring is recommended.
 - **Serum Albumin Levels:**
 - Lower normal range in neonates (2–4 g/dL) compared to adults; normalizes after 1 year.

- Highly protein-bound drugs (e.g., sulfamethoxazole-trimethoprim, ceftriaxone) not used due to risk of bilirubin displacement and kernicterus.
- **Lipid-Soluble Drugs:**
 - Despite lower body adipose in neonates, Vd for lipid-soluble drugs (e.g., lorazepam) is similar to infants and adults.
- **Blood–Brain Barrier:**
 - Immature in neonates, leading to higher CNS concentrations of some drugs (e.g., vancomycin, phenobarbital).

Metabolism

- **Hepatic Metabolism in Neonates:**
 - Slower at birth in full-term infants; delayed further in premature neonates.
 - **Phase 1 Reactions:**
 - Impaired (e.g., oxidation, alcohol dehydrogenase).
 - Ethanol-containing products should be avoided to prevent toxicities (e.g., respiratory depression).
- **Cytochrome P450 Enzymes:**
 - Activity reaches adult levels at variable ages depending on the isoenzyme.
 - Increased dose requirements for some drugs (e.g., phenytoin, valproic acid) in children aged 2–4 years due to higher liver-to-body mass ratio.
- **Phase 2 Reactions:**
 - **Sulfate Conjugation:** Well-developed at birth.
 - **Glucuronidation:** Immature in neonates; matures by 2–4 years.
 - Deficiency can cause "gray baby syndrome" (e.g., chloramphenicol toxicity).
- **Toxicity from Immature Conjugation:**
 - **Benzyl Alcohol/Benzoic Acid:** Avoid in neonates to prevent "gasping syndrome" (e.g., respiratory depression, seizures).
 - **Propylene Glycol:** Avoid due to risks of hyperosmolality, seizures, and respiratory distress.
- **Acetylation:**
 - Reaches adult levels at ~1 year; activity depends on genetic variability.

Elimination

- **Nephrogenesis and GFR Development**
 - Nephrogenesis completes at ~36 weeks gestation.
 - Premature neonates and infants have reduced glomerular and tubular function, leading to lower GFR.
 - Reduced GFR in neonates necessitates longer dosing intervals for renally cleared drugs (e.g., vancomycin).
- **Age-Related GFR Changes**
 - GFR increases with age, exceeding adult values in early childhood.
 - Gradually declines to adult levels during adolescence.
- **Vancomycin Dosing by Age Group**

- Low-birth-weight (LBW) premature neonates: every 18–24 hours.
- Children with normal renal function: every 6 hours.
- Adults with normal renal function: every 8–12 hours.
- **Cystic Fibrosis and Drug Clearance**
 - Children with cystic fibrosis show higher renal drug clearance (e.g., aminoglycosides).
 - Requires higher doses and more frequent dosing intervals compared to non-CF children.
- **Creatinine Clearance as GFR Marker**
 - Creatinine clearance is a surrogate for GFR.
 - Pediatric GFR calculated in mL/min/1.73 m².
- **Limitations of Adult GFR Estimation Equations**
 - Equations like Cockcroft-Gault, Jelliffe, CKD-EPI, and MDRD are unsuitable for patients under 18 years.
 - These equations can overestimate GFR or lack validation for pediatric use.
 - Example: Cockcroft-Gault poorly agrees with measured GFR in adolescents as young as 13 years.
- **Pediatric GFR Estimation**
 - **Schwartz Equation**
 - Historically used from infancy to 21 years.
 - Uses length (cm), serum creatinine (mg/dL or $\mu\text{mol/L} \times 0.0113$), and a constant (k) based on age/weight.
 - No longer widely used due to overestimation from IDMS-calibrated creatinine methods.
 - Preferred Method – "Bedside" Schwartz equation for children aged 1–18 years.
 - Compatible with IDMS-traceable creatinine methods, providing accurate GFR estimation.

FIGURE 2–3.

“Original” Schwartz equation for estimation of glomerular filtration rate (GFR) in pediatric patients up to 21 years of age. (From Schwartz GJ, Brion LP, Spitzer A. The use of plasma creatinine concentration for estimating glomerular filtration rate in infants, children, and adolescents. *Pediatr Clin North Am.* 1987;34(3):571–590.)

“Original Schwartz” Equation

$$\text{GFR} = kL/\text{SCr}$$

Age	K
Low birth weight < 1 year	0.33
Full term < 1 year	0.45
1–12 years	0.55
13–21 years (female)	0.55
13–21 years (male)	0.70

K = proportionality constant

L = length in cm

SCr = serum creatinine in mg/dL

GFR = estimated glomerular filtration rate in mL/min/1.73 m²

Source: M.A. Chisholm-Burns, T.L. Schwinghammer, P.M. Malone, J.M. Kolesar, K.C. Lee, P.B. Bookstaver: Pharmacotherapy Principles & Practice, 6th Edition. Copyright © McGraw Hill Education. All rights reserved.

FIGURE 2-4.

Bedside Schwartz equation for estimation of glomerular filtration rate (GFR) in pediatric patients ages 1 to 18 years old. (From Staples A, LeBlong R, Watkins CW, Brandt J. Validation of the revised Schwartz estimating equation in a predominantly non-CKD population. *Pediatr Nephrol*. 2010;25:2321–2326.)

“Bedside Schwartz” Equation

$$\text{GFR} = [0.413 \times h]/\text{SCr}$$

H = height in cm

SCr = serum creatinine (in mg/dL)

GFR = estimated glomerular filtration rate in mL/min/1.73 m²

Source: M.A. Chisholm-Burns, T.L. Schwinghammer, P.M. Malone, J.M. Kolesar, K.C. Lee, P.B. Bookstaver: Pharmacotherapy Principles & Practice, 6th Edition. Copyright © McGraw Hill Education. All rights reserved.

- Urine output is also a parameter used to assess renal function in pediatric patients.
- Infants and children with acute kidney injury are likely to present with oliguria or anuria with urine output less than 1 mL/kg/hour.

Specific Considerations in Drug Therapy in Pediatrics

- **Off-Label Medication Use**
 - Many drugs lack pediatric-specific dosing, safety, and efficacy data.
 - Off-label use involves indications, age ranges, dosing, or routes outside approved labeling.
 - Refer to studies for safety, efficacy, and dosing when using off-label medications.
 - FDA incentives promote pediatric drug studies, excluding generics.
- **Routes of Administration and Drug Formulations**
 - Route selection depends on age, disease, and severity.
 - Rectal: Reserved for specific cases.
 - Topical: Common for dermatologic conditions.
 - Injectable: For severe illnesses or when other routes are unsuitable.
 - Oral formulations:
 - Liquid forms preferred for children under 6 years.
 - Compounding may be required when commercial liquid forms are unavailable.
 - Factors to consider: stability, dose uniformity, palatability.
 - Avoid using extended-release formulations for crushed or mixed doses.
 - Flavor pairing (e.g., clindamycin with chocolate) improves palatability.
 - Educate caregivers on compounding pharmacy options and administration techniques.
- **Common Errors in Pediatric Drug Therapy**
 - Errors occur in up to 27% of pediatric prescriptions.
 - Key causes:
 - Incorrect dose calculation/preparation.
 - Inaccurate weight, height, and age documentation.
 - Decimal errors (e.g., trailing zeroes).
 - Look-alike labels and combination product dosing confusion.
 - Prevention strategies:
 - Use standardized concentrations and programmable infusion pumps.
 - Leverage clinical decision support systems in electronic health records.
 - Obtain complete medication histories and educate caregivers.
- **Complementary/Alternative Medicine (CAM) and OTC Use**
 - CAM use common in chronic illnesses, with limited efficacy data.
 - Risks include harmful additives and interactions with prescription drugs.
 - Example: Avoid aspirin in children with viral infections to prevent Reye syndrome.
 - Educate caregivers on safe OTC use and consult pediatricians for guidance.
- **Medication Administration and Caregiver Education**
 - Challenges: cooperation from infants/children, socioeconomic barriers, health literacy.
 - Recommendations:
 - Round doses for easy measurement (except narrow therapeutic index drugs).
 - Use oral syringes over cups or spoons for accuracy.

- Standardize dosing units (e.g., milliliters).
- Educate caregivers:
 - Proper storage, preparation, administration techniques.
 - Expected outcomes and adherence importance.
 - Calming strategies for children during administration (e.g., toys, rewards).

Table 2–4 Helpful Tips for Medication Administration for Selected Dosage Forms⁴⁶

Dosage Form	Recommendations
Ophthalmic drops or ointment	• Wash hands thoroughly prior to administration • Position child laying down in supine position • Avoid contact of applicator tip to surfaces, including the eye • Drops should be placed in the pocket of the lower eyelid • Ointment strip should be placed along the pocket of the lower eyelid
Otic drops	• Wash hands thoroughly prior to administration • Position child laying down in prone position • Tilt head to expose treated ear, gently pull outer ear outward, then due to age-dependent change in angle of eustachian tube: ○ If child < 3 years of age, gently pull downward and back; apply drops ○ If child is > 3 years of age, gently pull upward and back; apply drops
Nasal drops	• Wash hands thoroughly prior to administration • Position child laying down in supine position • Slightly tilt head back; place drops in nostril(s) • Remain in position for appropriate distribution of medication
Rectal suppository	• Similar to adult administration; challenging route for administration • For younger patients (ie, < 3 years), a smaller finger (eg, pinky finger) should be used to insert suppository
Metered-dose inhalers	• Use a spacer ○ For younger children, use one with a mask, be sure the mask is secured/placed closely up against the child's face, avoiding gaps between face and mask and creating a seal to ensure medication delivery • Child should take slow breaths in with each dose • Wait at least 1 minute between doses

Patient Encounter Part 3

DD is a 13-year-old adolescent boy who is brought to the clinic with a weeklong history of productive cough, rhinorrhea, and congestion. The child's temperature last night was 39°C

(102.2°F) by electronic axillary thermometer. Mom reports to have given the child several doses of acetaminophen, but symptoms did not improve and thus she brought him to be seen. Patient has history of allergic rhinitis and eczema. The pediatrician diagnoses DD with community-acquired pneumonia requiring antibiotic treatment and asks you to develop a treatment care plan for DD including use of cefdinir 14 mg/kg/day by mouth divided twice a day for 10 days.

The medical assistant provides you DD's weight as 85.8 lb (39 kg). Mom states that DD would prefer a liquid form as he is still working on swallowing pills. Cefdinir is available as a 125 mg/5 mL or 250 mg /5 mL suspension. What is the most appropriate dosage strength and calculated dose of cefdinir for DD?

Two days later, Mom calls the clinic stating that DD did not like the taste of cefdinir and would like to change to a capsule or tablet. Cefdinir comes as a 300-mg capsule. Explain whether or not this change is appropriate.